What is the File Bridge

The File Bridge allows for Exemplar to communicate with file systems on your local network in a cryptographically secure fashion.

How to Use

After setup to use the file bridge at supported locations all you need to do is change the path present to one that matches the following format bridge://conn/path_on_local.

An example:

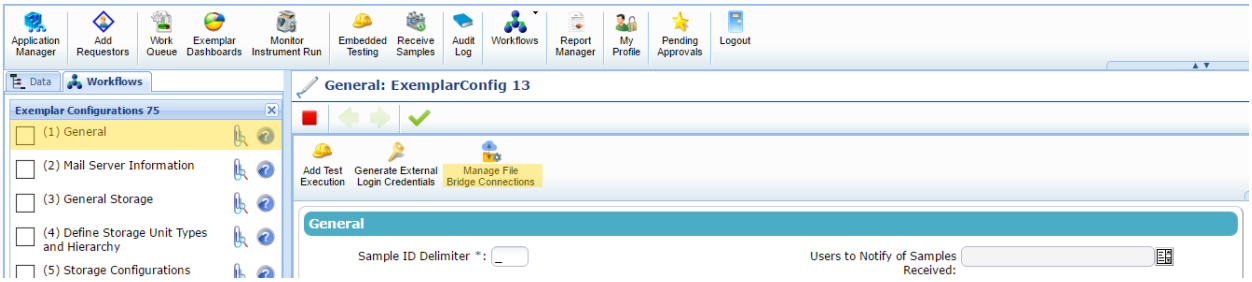
- bridge://alpha/instrument_412/output
    - This will find the connection with a name matching alpha and navigate to the %repository_root%/instrument_412/ouput directory on the repository side.

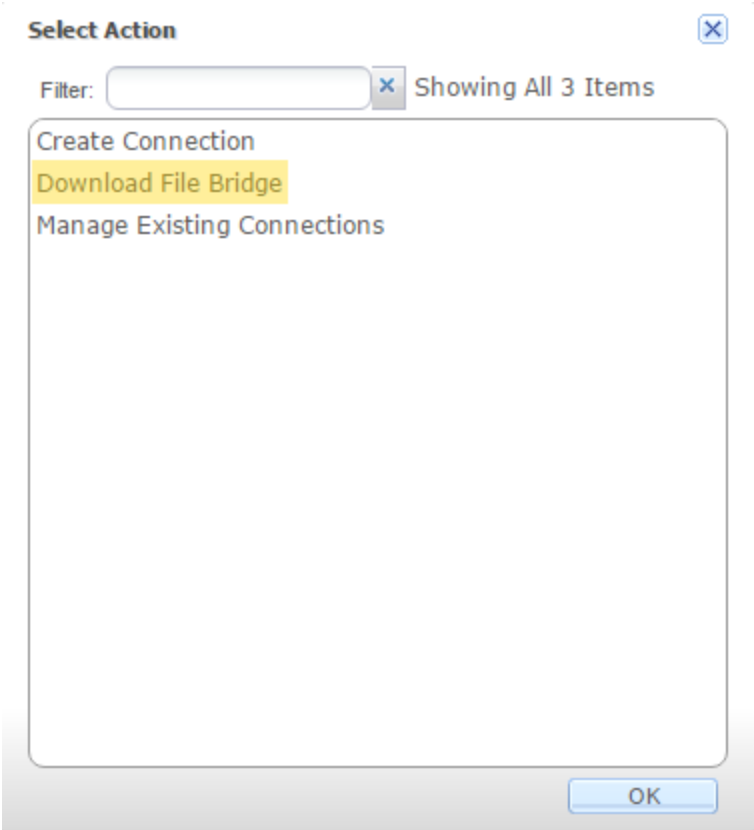
Exemplar Installation

Before proceeding with installing the File Bridge a working LIMS with the Exemplar Configurations WF and the file bridge plugin must be present. If you're using our hosted solution, this should be done for you already.

File Bridge Installation

You can manage the File Bridge on the Exemplar side from the Manage File Bridge Connections button present on the Exemplar Configurations/General task.

1. 

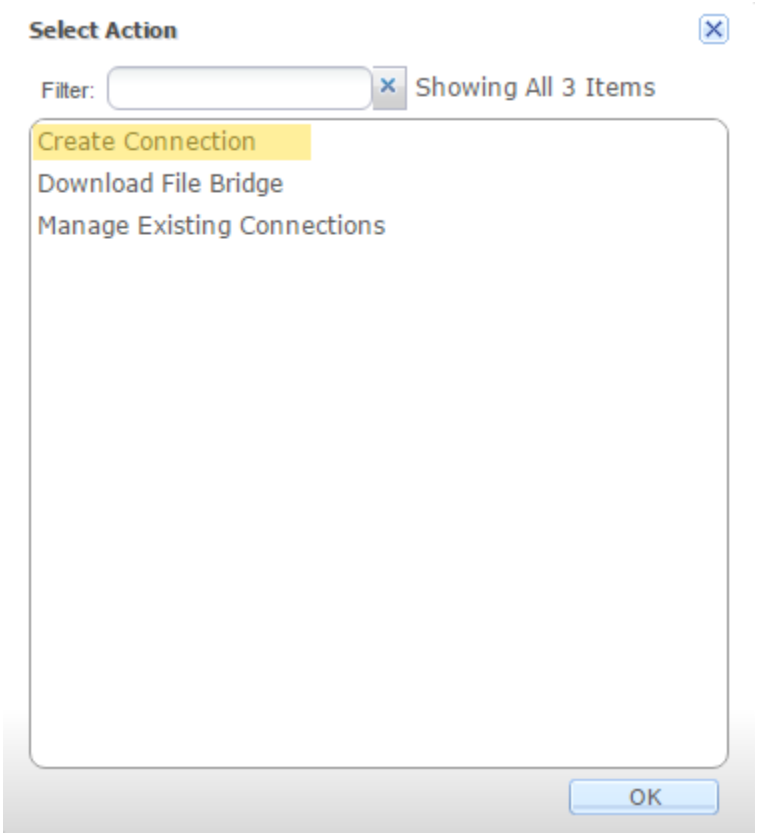
2. 

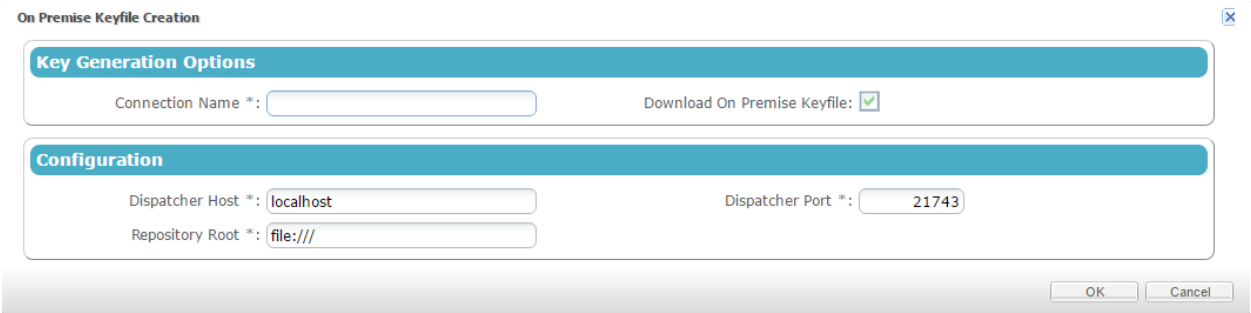
   Explanations of the options present here

   a. Create Connection
      i. This will allow for you to define new File Bridge connections to be used in Exemplar.
   b. Download File Bridge
      i. This allows you to download the file bridge agent, which will connect to the proxies defined in the Create Connection step.
   c. Manage Existing Connections
      i. This option provides a screen for selecting connections to download/delete. If only one connection is selected it allows you to browse the hosted file system to verify that you can indeed access it.

3. After you download the file bridge you can extract it where desired (/opt/sapiosciences/filebridge/ is the preferred location and where the linux service script settings default to).
   a. After you extract the file bridge copy the filebridge service script from service/linux/filebridge to /etc/init.d/ if applicable for your distro
      i. You may need to configure this script for your installation
      ii. This service script gives start/stop/restart commands
   b. If desired set the service to automatically run on boot with chkconfig –add filebridge

4. Now that the repository host is extracted & installed we can add a connection for Exemplar and the repository to establish a connection to each other.





   a. Name.
      i. This is how you'll reference the connection in paths. This can't be changed after the key is generated. If you want to change this you'll have to delete the old key and create a new one with the name desired and update the repository accordingly
   b. Proxy Host & Port
      i. This is what both Exemplar and the Agent Host will connect to in order to transfer data. If you're using our hosted solution we'll provide you with a host & port that should be used. If you're hosting Exemplar you'll need to configure the Proxy as well.

c. Agent Root
    i. For those familiar with chroot/jailing this essentially accomplishes the same functionality. The root path set here will become the root for paths that reference this connection. If access is made outside the root it'll throw an error to the caller.

5. Upon creation the connection will be stored in Exemplar for future use and a copy will be sent to your browser to download.
6. You'll copy the downloaded zip to the connections folder without extracting it. Upon running the file bridge it will scan this directory and read the connection details in all the zips present

Programmatic API

The API is designed to be easy to **use** and the standard way of accessing File Bridge Repositories is with the Java 7 NIO.2 Path & FileSystem interface. By using the NIO.2 interface it allows for developers to support both local and remote filesystems in the present and the future without complex file logic.

How to use:

- Path path = SapioPaths.get("bridge://alpha/instrument_412/output");
    - We use our own path resolver as the default Paths resolver requires that the classes be in the system classloader which doesn't work with the Exemplar plugin paradigm. SapioPaths.get will also resolve local file:// references as well.
- General Pathing
    - Path child = path.resolve("child");
        - Resolves to: bridge://alpha/instrument_412/output/child
    - Path parent = path.getParent();
        - Resolves to: bridge://alpha/instrument_412/
- File I/O
    - byte[] bytes = Files.readAllBytes(path)
        - Will read all the bytes from the given path and return it.
    - Files.write(path, bytes)
        - Will write all the bytes to the given path
    - Files.delete(path)
        - Deletes the file/directory at the end of the path
    - InputStream is = Files.newInputStream(path)
        - Opens and returns an InputStream for the given path.
    - OutputStream os = Files.newOutputStream(path)
        - Opens and returns an OutputStream for the given path.
    - Files.createDirectories(path)
        - Attempts to create any missing directories in the given path
    - Files.createDirectory(path)

        - Attempts to create a single directory at the end of the path
    - Files.createFile(path)
        - Attempts to create a single empty file at the end of the path